

Practice Questions

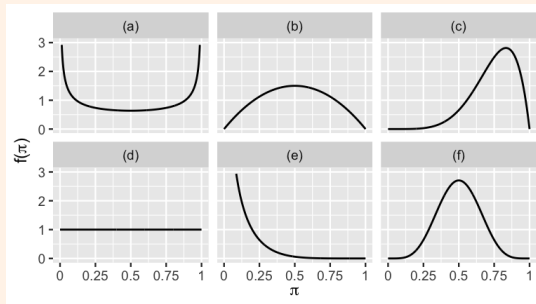
1. If prior concentration doubles but its mean stays fixed, what changes?
2. Why is $L(\pi \mid y)$ not $P(\pi \mid y)$?
3. What does conjugacy buy us computationally?
4. When can two analysts share a posterior but rationally choose different actions?
5. Which assumption would you challenge first in the election-poll example?
6. Identify the most decision-sensitive assumption.
7. Provide one example where false-positive and false-negative costs differ.

Practice Questions

8. Six Beta pdfs are plotted below.

Match each to one of the following models:

Beta(0.5,0.5), Beta(1,1), Beta(2,2), Beta(6,6), Beta(6,2), Beta(0.5,6).



9. What do you call a sweet carbonated drink: pop, soda, coke, or something else? Let π be the proportion of U.S. residents that prefer the term “pop.” Two different beverage salespeople from different regions of the country have different priors for π . The first salesperson works in North Dakota and specifies a Beta(8,2) prior. The second works in Louisiana and specifies a Beta(1,20) prior.
- Calculate the prior mean, mode, and standard deviation of π for both salespeople.
 - Plot the prior PDFs for both salespeople.
 - Compare, in words, the salespeople’s prior beliefs about the proportion of U.S. residents who say “pop.”

10. A university wants to know what proportion of students are regular bike riders, π , so that they can install an appropriate number of bike racks. Since the university is in sunny Texas, staff think that π has a mean of 1 in 4 students, and a mode of 5/22.
- a. Specify and plot a Beta model that reflects the staff's prior ideas about π .
 - b. Among 50 surveyed students, 15 are regular bike riders. What is the posterior model for π ?
 - c. What are the mean, mode, and standard deviation of the posterior model?
 - d. Does the posterior model more closely reflect the prior information or the data? Explain your reasoning.